

Herbal monoterpene alcohols inhibit propofol metabolism and prolong anesthesia time

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Abstract

2,6-Diisopropylphenol (Propofol) is a short-acting intravenous anesthetic that is rapidly metabolized by glucuronidation and ring hydroxylation catalyzed by cytochrome P450. The goal of this research was to determine whether dietary monoterpene alcohols (MAs) could be used to prolong the anesthetic effect of propofol by inhibiting propofol metabolism in animals. Mice were injected intraperitoneally (i.p.) with MAs (100–200 mg/kg) followed by the administration of 100 mg/kg propofol 40 min later via an i.p. injection. The time of the anesthesia of each mouse was recorded. It was found that (±)-borneol, (–)-carveol, *trans*-sobrerol, and menthol significantly extended the anesthetic effect of propofol (>3 times). The concentration of propofol in the mouse blood over time (up to 180 min) also increased in mice pre-treated with (–)-borneol, (–)-carveol, and *trans*-sobrerol. The volume of distribution of propofol decreased in the (–)-borneol ($p < 0.05$), pre-treated group as compared to the propofol control group. Moreover, the maximum blood concentration of propofol and the concentration of propofol in the blood as indicated by the area under the curve were significantly increased in (–)-borneol and (–)-carveol pre-treated groups. Additional evidence using rat hepatocytes showed that (–)-borneol inhibited propofol glucuronidation whereas *trans*-sobrerol and (–)-carveol inhibited cytochrome P450 dependent microsomal aminopyrine *N*-demethylation. These results suggest that (–)-borneol extends propofol-induced anesthesia by inhibiting its glucuronidation in the mouse whereas *trans*-sobrerol (–)-carveol extends propofol-induced anesthesia by inhibiting P450 catalyzed propofol metabolism.

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Keywords: Monoterpene alcohol; Propofol; Pharmacokinetics; Glucuronidation; P450

Introduction

2,6-Diisopropylphenol (Propofol) is widely used as a short-acting intravenous anesthetic for a wide range of surgical procedures including general surgery, urology, and orthopaedic surgery (Trapani et al., 2000). The duration of the propofol anesthetic effect is approximately 5–10 min following a single dose of 1.5–2.5 mg/kg intravenously. This short duration of action is due to its rapid redistribution and metabolism by glucuronidation (Roschau and Roschlau, 1998). As shown in

Fig. 1, propofol undergoes rapid biotransformation to various metabolites in human liver: 53–72% through glucuronidation (catalyzed by uridine diphosphate-glucuronosyltransferases (UGT) such as UGT1A9); and ring hydroxylation to 4-hydroxypropofol (4-HP) catalyzed by the P450 isozymes CYP2C and CYP2B (Court et al., 2001; Le Guellec et al., 1995). 4-HP then undergoes glucuronidation and sulphation to form 1-(4-HP)-glucuronide (1-QG), 4-(4-HP)-glucuronide (4-QG), and 4-(4-HP)-sulphate (4-QS) (Court et al., 2001; Guitton et al., 1998; Sneyd et al., 1994). Rabbit and rat liver microsomes catalyse propofol glucuronidation less rapidly than human liver microsomes (Le Guellec et al., 1995).

Monoterpene alcohols (MA) are found in the volatile essences of flowers, oils of various plants, and herbal medicines. Some are commonly used as food additives and fragrance components and are found in cosmetics, soaps, and cleaning products (Table 1) (Guitton et al., 1998). (–)-Borneol (Fig. 2) is

Abbreviations: MA, monoterpene alcohols; UDPGA, uridine diphosphate-glucuronic acid; i.p., intraperitoneally; C_{max} , maximum propofol concentration; V_{ss} , volume of distribution at steady state; AUC, area under the curve; CL_T , total clearance; MRT, mean residence time; UGT, uridine diphosphate-glucuronosyltransferase; SULT, sulfotransferases; CYP, cytochrome P450 isozymes.

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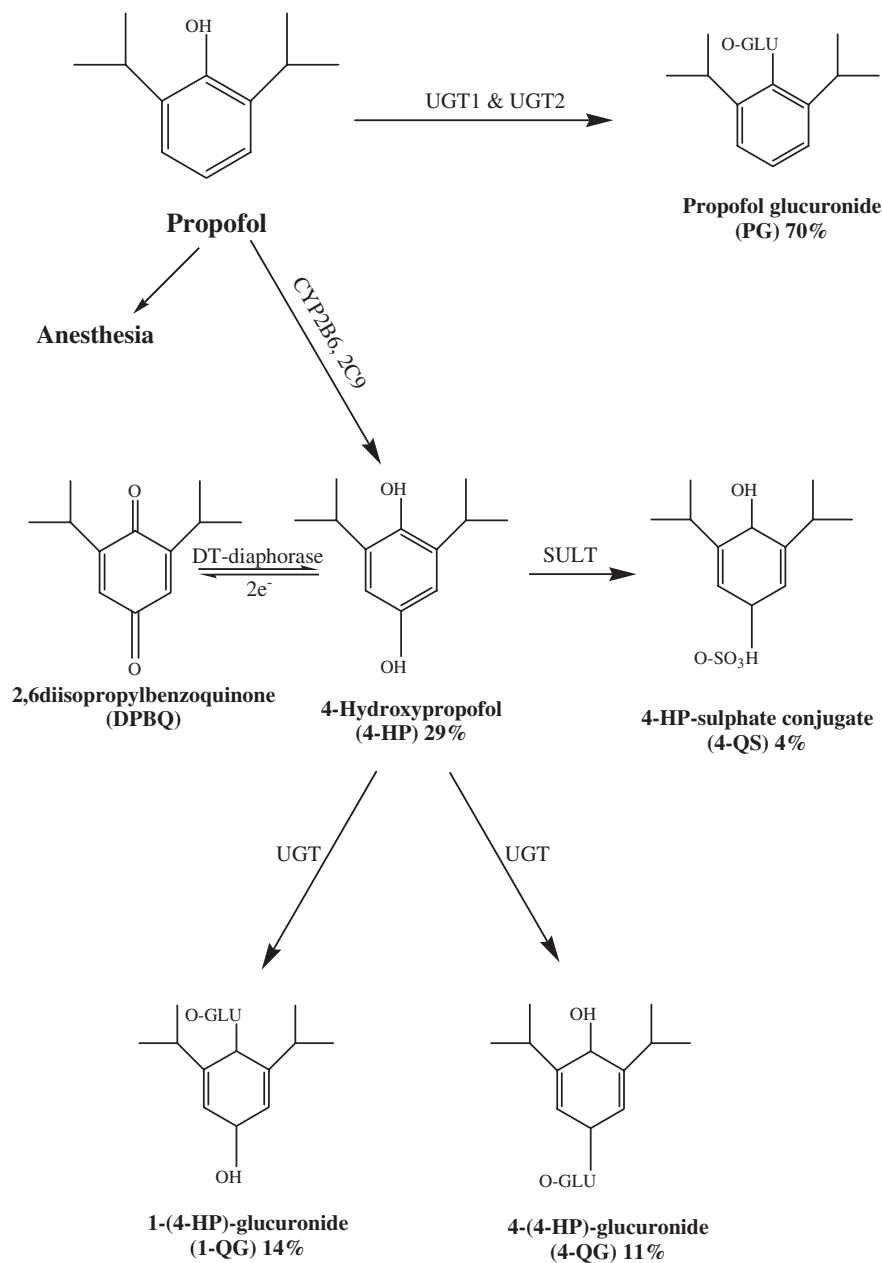


Fig. 1. The metabolic pathway of propofol in humans. UGT — uridine diphosphate-glucuronosyltransferase, SULT — sulfotransferases, CYP — cytochrome P450 isozymes. Percent of metabolites in urine as given.

often the main constituent of the essential oils of medicinal herbs such as the genus *Micromeria* (Sneyd et al., 1994), *Pinus* (B.E. A.L. Buchanan, 2000), and rosemary (Tabanca et al., 2001). It has also been reported to have antibacterial, antifungal, antispasmodic, choleric, and tranquilizing effects (Angioni et al., 2004). Borneol at 320 mg/kg administered i.p. to male SD rats was recently shown to prevent liver glucuronidation in vivo for nearly 2 h (Angioni et al., 2004; Tabanca et al., 2001) probably as a result of inhibiting hepatic UGT and depleting hepatic UDPGA by 85–90%. Menthol is the major component of peppermint and thyme (Tabanca et al., 2001). It is used as a common cold remedy (Tabanca et al., 2001) and has antibacterial effects. Menthol also inhibits CYP2A6 and CYP2B (Li et al., 2003; Tabanca et al., 2001). *trans*-

Sobrerol is another type of monoterpene alcohol that is widely used as a mucoregulating agent for the treatment of respiratory diseases with bronchial obstruction and mucus hypersecretion (Li et al., 2003; Watkins and Klaassen, 1983). Ventura et al. has found that *trans*-sobrerol undergoes a hepatic P450 catalysed hydroxylation, in vivo in rat, dog, and human (De Oliveira et al., 1999; Pierce, 1999; Ransom and Mueller, 1997).

Propofol itself does not produce complete anesthesia because it lacks analgesic effects and has a short anesthetic effect after a single bolus administration (MacDougall et al., 2003; Ventura et al., 1983; Ransom and Mueller, 1997). Therefore, it is commonly used clinically in combination with opioids or inhaled anesthetics in order to produce a complete anesthesia

Table 1
Terpenoids found in herb/species/fruits

Terpenoids	Herbs/species
Carvone/carveol	Caraway, spearmint, and dill
Cineole	Coriander, lavender, rosemary, sage, and thyme
<i>trans</i> -Sobrerol	Sage
Borneol	Rosemary type B
Farnesol	Lemongrass, chamomile, and lavender
Geraniol	Lemongrass, coriander, melissa, basil, and rosemary
Limonene	Caraway, mints, cardamom, dill, celery seed, coriander, and fennel
Menthol	Peppermint and thyme
Perillyl alcohol	Lavender, spearmint, sage
α -pinene	Caraway, coriander, fennel, juniper berry, rosemary, and thyme

state. In the following, we have focused on determining whether the anesthetic effectiveness of propofol is affected by inhibiting hepatic glucuronidation by the previous administration of borneol (an MA) or by inhibiting hepatic P450 isozymes with menthol (another MA). Such studies may help address the issue of whether herbal medicines could modulate propofol anesthesia? Also, the effects of pre-treatment with MAs on the pharmacokinetics of propofol have been studied. It was found that the anesthetic effect of propofol and blood propofol concentrations was also increased when mice were pre-treated with these MAs.

Materials and methods

Chemicals

2,6-Diisopropylphenol (propofol), propylene glycol, ($\pm$)-borneol, α -terpineol, menthol, *trans*-sobrerol, (–)-carveol, metaphosphoric acid (MPA), D-saccharic-1,4-lactone monohy-

drate, dimethyl sulfoxide (DMSO), sodium bicarbonate, Biorad protein assay reagent (Biorad #500-0006), magnesium chloride (MgCl_2), aminopyrine, nicotinamide adenine dinucleotide phosphate; oxidized form (NADP^+), glucose-6-phosphate, glucose-6-phosphate dehydrogenase, potassium phosphate, uridine diphosphate-glucuronic acid (UDPGA), sodium deoxycholate, trichloroacetic acid (TCA), hepes, and bovine serum albumin were purchased from Sigma-Aldrich Chemical Company. Acetonitrile, glacial acetic acid, and methanol were EMD grade from OmniSolv solvents.

Animals

8 week old male, CD-1 mice (30–40 g) and adult male Sprague–Dawley rats (280–300 g) were purchased from Charles River Company (St. Constant, Quebec). Mice (3–4 per cage) and rats were housed (1–2 per cage) with free access to water and food (*ad libitum*) in the animal care facility for at least one week before study, under a 12 h light/dark cycle.

Methods

In vivo studies

Assessment of the anesthetic effects of propofol. The mice ($n=25$) were divided into 5 groups. Group 1 was injected with vehicle (propylene glycol) intraperitoneally (*i.p.*) and groups 2 to 5 were injected via *i.p.* with propofol at doses of 50, 100, 250, and 500 mg/kg. The anesthetic effects were evaluated in mice following propofol administration. The time of the analgesic effect was measured when the mice did not right themselves within 30 s after being placed in a supine position in accordance with the method of Fassoulaki et al. (1994).

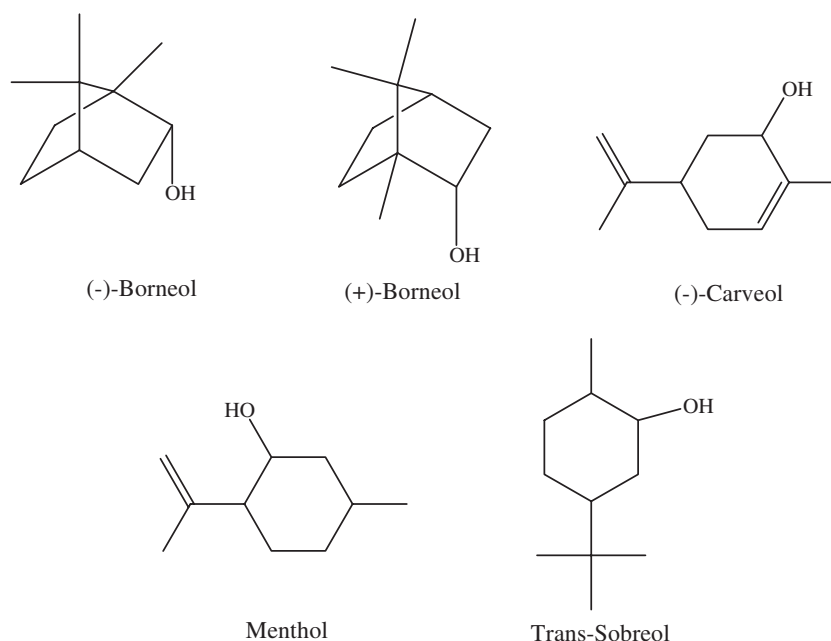


Fig. 2. The chemical structures of isoterpene and monoterpene alcohols.

In a separate study, mice ($n=35$) were divided into 6 groups. Group 1 ($n=4$) was treated with propofol (100 mg/kg) alone. The other five groups of mice were pre-treated i.p. with 200 mg/kg of (–)-carveol, menthol, (+)-borneol, (–)-borneol, and *trans*-sobrerol 40 min before propofol administration (100 mg/kg). The anesthetic effects were evaluated as mentioned above following propofol administration.

HPLC analysis of propofol concentration in the mouse blood
Blood samples preparation. Blood propofol concentrations were measured after the mice were pre-treated with 100 mg/kg (–)-borneol, (–)-carveol, and *trans*-sobrerol. The method of blood preparation was modified from Knibbe's method (Knibbe et al., 1998). Approximately 1 ml blood ($n=4$ /time point) was collected at each time point (0, 10, 20, 30, 60, 90, 120, and 180 min) by cardiac method. 50 μ l blood was aliquot into Eppendorf tubes, immediately hemolysed in 250 μ l millipore water, and kept at 4 °C (stable for about a month).

Methanol (20 μ l) was added to these 300 μ l blood samples and vortexed for 30 s. 500 μ l acetonitrile was added to each blood sample and vortexed for 2 min followed by 5 min of sonication. The mixture was then centrifuged for 5 min at 3000 $\times g$. 150 μ l of supernatant from each sample was analyzed by HPLC. The propofol concentration was found to be directly proportional to the peak height; therefore, propofol was calculated by measuring the height of the peak according to the propofol standard curve. The concentration of propofol in the blood was calculated using standards prepared in untreated blood.

Standard preparation. The stock concentrations of propofol used were 2, 10, 20, 30, 50, 70, 100, 180, 250 μ g/ml (in methanol). 20 μ l of each standard stock solution was added to control blood (300 μ l) and vortexed about 30 s. 500 μ l acetonitrile was added to each blood sample, vortexed for 2 min and sonicated for 5 min. The samples were then centrifuged for 5 min at 3000 $\times g$. Finally, 150 μ l supernatant of each sample was injected into the HPLC column. The concentrations of blood standard calculated before hemolysis in distilled water; corresponded to 0.8, 4, 8, 12, 20, 28, 40, 72, 100 μ g/ml. The standard was plotted based on the height of the peak versus propofol concentration.

Blood propofol analysis by HPLC. Propofol was detected by reverse phase liquid chromatography 13–18 using a Synergi C18 (150 $\times$ 4.6 mm) column (Phenomenex Inc.) connected to a Shimadzu SCL-6B system controller and a Shimadzu SPD-6AC UV–VIS spectrophotometer. The wavelength was set at 270 nm, and the signal was recorded using Shimadzu C-R3A integrator. The mobile phase for measuring propofol in the blood consisted of a mixture of acetonitrile, millipore water, and glacial acetic acid (60:40:0.1, v/v/v). The flow rate was 1.5 ml/min and propofol in the blood sample was eluted at 7.3 min.

In vitro studies

Isolation and incubation of rat hepatocytes. Hepatocytes were isolated by collagenase perfusion of the liver as previously

described by Moldeus et al. (1978). Cells (10^6 hepatocytes/ml) were suspended in round bottomed flasks continuously rotating in a water bath at 37 °C in Krebs–Henseleit Buffer (pH 7.4), supplemented with 12.5 mM Hepes in a gaseous atmosphere of 95%O₂/5%CO₂. Cell viability was determined by measuring the exclusion of trypan blue (final concentration: 0.2% w/v). Hepatocytes were preincubated for 30 min in the water bath at 37 °C before the addition of test chemicals. Modulators were used at non-toxic concentrations and preincubated with hepatocytes before propofol. The preincubation time of each modulator was specified in each table and figure.

(–)-Carveol, *trans*-sobrerol, and (–)-borneol were used at non-toxic concentrations and were incubated for 30 min with the hepatocytes prior to propofol (100 μ M) addition. A 1 ml aliquot from each round-bottom flask (containing 1×10^6 hepatocytes) was transferred to an Eppendorf tube at time 30, 90, and 150 min. Then, 100 μ l metaphosphoric acid (15% w/v) was added to each sample to precipitate protein and the samples were centrifuged at 13500 $\times g$ for 5 min. A 100 μ l aliquot of the supernatant was injected onto the column. Other samples were kept at –20 °C until analysis. Propofol glucuronide (PG) was separated by HPLC.

Assay of microsomal and hepatocytes propofol glucuronidation. Rat liver microsomes were prepared as previously described (Dallner, 1978). The protein content was determined using a Biorad protein assay provided by the company. Microsomes (50 μ g/ml), UDPGA (1 mM) and D-Saccharic-1,4-lactone monohydrate (0.4 mM, β -glucuronidase inhibitor), were incubated with propofol (200 μ M) in the absence or presence of (–)-borneol (200 μ M), (–)-carveol (200 μ M), or *trans*-sobrerol (200 μ M) in 1 ml incubation buffer [containing potassium phosphate buffer (0.1 M, pH 7.4), 1% sodium deoxycholate (5% w/v), and 15 mM MgCl₂] at 37 °C for 2 h. The reaction was terminated by adding 100 μ l of MPA (15% w/v) to precipitate protein. An aliquot of 100 μ l of the supernatant was injected into the HPLC column.

A 10 μ l aliquot of each standard propofol stock solution (1, 5, 10 mM in DMSO) was added to 1 ml control hepatocytes (1×10^6 hepatocytes) and incubated for 90 min. 100 μ l of MPA was added to 1 ml aliquot samples of the incubate and then centrifuged at 13,500 $\times g$ for 5 min. Finally, 100 μ l supernatant of each sample was injected onto a HPLC column. The final concentrations of propofol incubated with the hepatocytes were 10, 50 and 100 μ M.

The HPLC system used is described above under blood propofol analysis. The mobile phase for measuring propofol and its glucuronide in the microsomes consisted of two solvents: solvent A, water and glacial acetic acid (pH 3.8); solvent B: acetonitrile. Elution was performed with a linear gradient profile where solvent B varied from 13% to 89% in 35 min. The flow rate was 0.8 ml/min. The retention time of propofol and propofol glucuronide were 29.3 and 20.3 min, respectively. Propofol glucuronide and propofol were verified by liquid chromatography/mass spectrometry (LC/MS) using a PSIEX bio-mass spectrometer (in positive mode). The conditions of the LC mobile phase and column were the same as mentioned

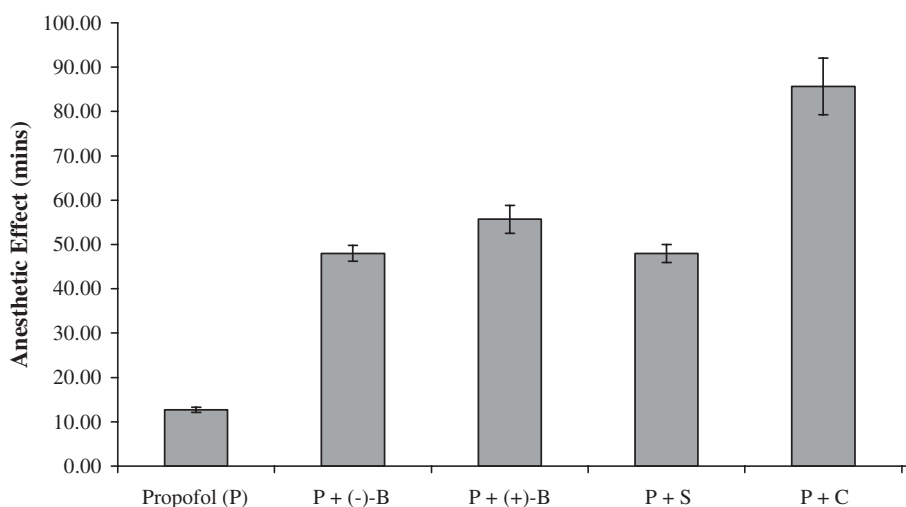


Fig. 3. The anesthetic effect of propofol was prolonged by monoterpene alcohols. Note: each value was expressed as mean $\pm$ S.E. of 4–6 mice. The dose of propofol administered via i.p. was 100 mg/kg. The doses of the other monoterpene alcohols used were 200 mg/kg. Propofol — P, (+)-borneol — (+)-B, (-)-borneol — (-)-B, *trans*-sobrerol — S, (-)-carveol — C, menthol — M. All monoterpene alcohols showed a significant extension of the anesthetic effect of propofol ($P < 0.001$) as compared with control propofol. Note: all of the monoterpene alcohols alone did not induce any anesthetic effect.

above for rat microsomes. The percent of propofol glucuronide formation in (-)-borneol, (-)-carveol, and *trans*-sobrerol pretreated groups were measured based on the peak height of propofol glucuronide in the control group.

Assay of microsomal aminopyrine-N-demethylase activity.

Microsomal aminopyrine N-demethylase activity was assayed using an incubate containing 0.1 M TRIS-HCl buffer (pH 7.4), 0.3 mg/ml rat liver microsomal protein, NADPH-generating system (1 mM NADP⁺, 2.5 mM glucose-6-phosphate, 0.4 U/ml glucose-6-phosphate dehydrogenase, 2.5 mM MgCl₂), inhibitors (0.5 mM), and aminopyrine (10 mM) in a final volume of 2 ml. Inhibitors were dissolved in DMSO whereas aminopyrine was dissolved in water. The inhibitors were propofol, (-)-borneol, (-)-carveol, and *trans*-sobrerol and were added to the incubation mixtures 10 min before adding the NADPH generating system and 15 min before adding aminopyrine. The incubation mixtures were incubated at 37 °C for 2 h after aminopyrine was added. The reaction was stopped by adding 56 μ l of TCA (30%, w/v). The mixtures were centrifuged at 100 $\times$ g for 5 min and the supernatant (500 μ l) of each mixture was transferred to a new glass tube containing 500 μ l of NASH reagent (1 M acetylacetone, 4 M ammonium acetate and 1 M glacial acetic acid dissolved in H₂O). Then these mixtures were incubated at 37 °C for 40 min. The reaction was stopped by immersing the mixtures in the ice water and the mixtures were measured at 412 nm with a spectrophotometer. The aminopyrine N-demethylase activity was determined according to the amount of formaldehyde interacting with NASH reagent (NASH, 1953).

Statistical analysis

Statistical significance between control and treated groups in the present study was determined by one-way analysis of variance (ANOVA) test using SigmaStat 2.03.

Results

Dose-dependence of propofol anesthesia

The state of anesthesia was assessed by observing the loss of righting reflex within the first few minutes after propofol administration. The propofol-induced anesthetic effect was dose-dependent. Its anesthetic effect lasted 13 min at a dose of 100 mg/kg and 32 min at a dose of 250 mg/kg. A dose of 100 mg/kg was chosen to induce a short anesthetic effect. Some lethality

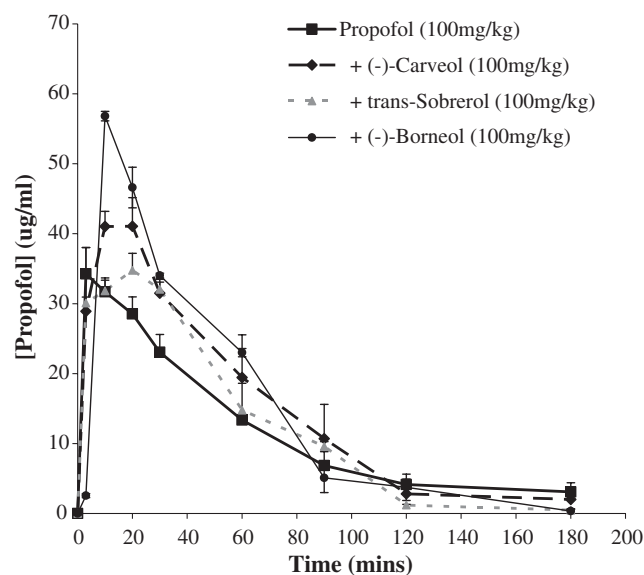


Fig. 4. The elimination of propofol in the blood over time after the mice were pretreated with (-)-borneol, (-)-carveol, and *trans*-sobrerol. P — mice treated with propofol; P+B — mice treated with propofol and (-)-borneol; P+C — a group of mice treated with propofol and (-)-carveol, P+S — mice treated with propofol and *trans*-sobrerol. The doses of all compounds were 100 mg/kg. Each value was expressed as mean $\pm$ S.E. from 3 to 4 mice. Propofol concentrations at each time interval were statistically analyzed by using one-way ANOVA test.

Table 2

Effect of (–)-borneol, *trans*-sobrerol and (–)-carveol on the pharmacokinetic parameters of propofol in the mouse blood using non-compartment model

Parameters	P	P+S	P+C	P+B
$T_{1/2\text{eff}}$ (min)	33.64±5.44	34.71±2.26	41.74±13	21.76±5.16
$T_{1/2\text{term}}$ (min)	140.8±53.55	—	311.00±86.21	—
CL_T (ml/min)	1.75±0.17	1.59±0.09	1.40±0.23	1.29±0.04
MRT (min)	46.38±2.99	38.65±3.55	51.70±2.64	46.19±3.59
V_{ss} (ml)	80.41±7.11	61.15±3.51	64.74±4.35	59.31±2.87*
C_{max} (μg/ml)	37.32±1.98	35.31±2.27	42.35±2.84	56.82±0.68**
$\text{AUC}_{0 \rightarrow \infty}$ (μg/ml/min)	2058.1±99.78	2323.87±45.40	2991.30±92.94**	2717.70±81.08*

P — mice treated with propofol; P+S — a group of mice treated with propofol and *trans*-sobrerol; P+C — mice treated with propofol and (–)-carveol; P+B — mice treated with propofol and (–)-borneol. $T_{1/2\text{eff}}$ — the effective half-life of propofol in the mouse blood; $T_{1/2\text{term}}$ — the terminal half-life of propofol in the mouse blood; CL_T — the rate of total propofol clearance; MRT — mean residence time; V_{ss} — volume of distribution at steady state; C_{max} — the peak concentration of propofol; AUC — area under the curve.

— Not measurable under the detection limit.

Each value was expressed as mean±S.E. of three mice.

*, ** The values were significantly different from P group ($P<0.05$, 0.001 respectively).

occurred with 500 mg/kg propofol administration, likely as a result of respiratory depression.

Extending the anesthetic effect of propofol in mice with monoterpene alcohol pretreatment

Selected MA were injected i.p. at doses 100–200 mg/kg to the mice 40 min before 100 mg/kg (561 μM) propofol administration. Forty minutes was the optimal preincubation time for MA to prolong propofol's anesthesia. The doses of MA used in this study were at least 10 times lower than their oral $\text{LD}_{50\text{s}}$ and by

themselves did not induce any anesthetic effect. All of the MAs tested significantly enhanced the anesthetic effect of a single dose of propofol by at least 3-fold (Fig. 3). The mice fully recovered from anesthesia approximately 6 h after waking.

The pharmacokinetics of propofol in the mouse blood after pretreatment with (–)-borneol, (–)-carveol, and *trans*-sobrerol

The effect of (–)-borneol, (–)-carveol, or *trans*-sobrerol (100 mg/kg, i.p.) on propofol metabolism was investigated by

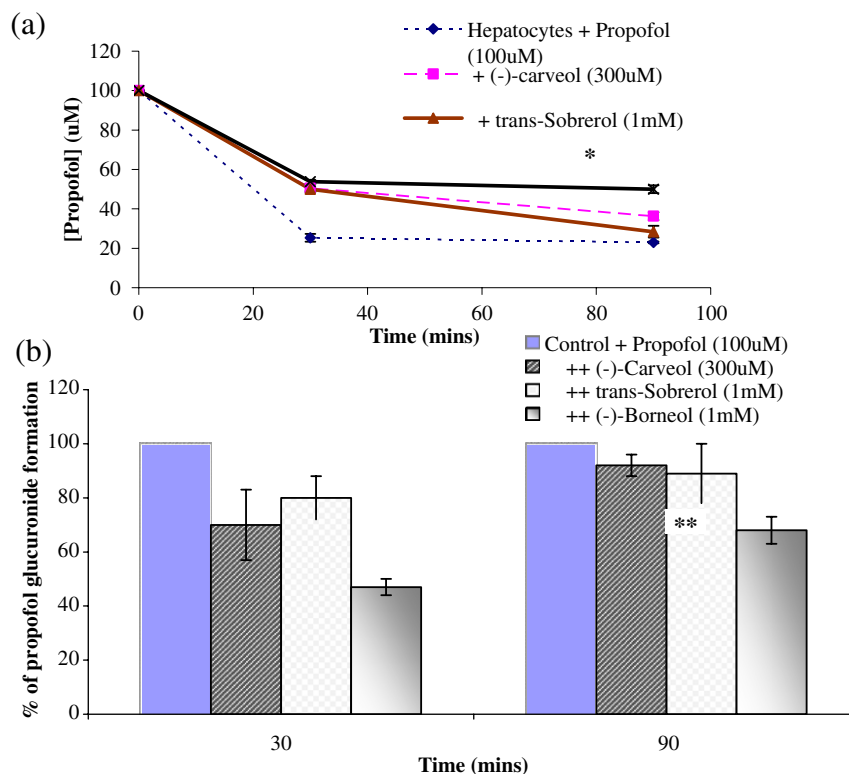


Fig. 5. Inhibition of propofol glucuronidation by monoterpene alcohol in isolated rat hepatocytes. (a) Total propofol metabolism. (b) Inhibition of hepatocytes propofol glucuronidation by preincubating hepatocytes with (–)-carveol, *trans*-sobrerol, and (–)-borneol for 30 min. Note: each value was expressed as mean±S.E. from 3 individual experiments. The values of the peak of PG are expressed as a percentage of the baseline value which is normalized to 100%. *, ** The value was significantly different from propofol ($P<0.05$, 0.001, respectively) using one-way ANOVA test.

Table 3

A) Microsomal propofol glucuronidation inhibition by monoterpene alcohols (100% propofol glucuronidation refers to a glucuronidation rate of 32 nmol/mg/h); B) microsomal cytochrome P450 dependent aminopyrine *N*-demethylation inhibition by propofol and monoterpene alcohols

A)	
Compounds added	% Propofol glucuronide formation
Propofol (200 μ M)	100
+(-)-Borneol (200 μ M)	49 $\pm$ 2**
+(-)-Carveol (200 μ M)	78 $\pm$ 4
+ <i>trans</i> -Sobrerol (200 μ M)	87 $\pm$ 6

B)	
Compounds added	HCHO formation (nmol/mg/min)
Aminopyrine (10 mM)	10.10 $\pm$ 0.50
+1-benzylimidazole (0.5 mM)	1.87 $\pm$ 0.07**
+(-)-Carveol (0.5 mM)	2.65 $\pm$ 0.26**
+ <i>trans</i> -Sobrerol (0.5 mM)	5.87 $\pm$ 0.36**
+Menthol (0.5 mM)	6.33 $\pm$ 0.36**
+(-)-Borneol (0.5 mM)	9.22 $\pm$ 0.27

The monoterpene alcohols were preincubated with microsomes 15 min before aminopyrine addition.

Each value was expressed as mean $\pm$ S.E. of 3 separate experiments.

The values with symbol * and ** were significantly different than propofol control ($P<0.001$ respectively).

measuring changes in the concentration of propofol mouse blood. After administration, propofol was significantly higher in the pretreated animals than in the propofol control group (Fig. 4). All groups pre-treated with (-)-borneol and (-)-carveol had higher blood-propofol concentrations than the propofol control group for up to 60 min after propofol administration ($p<0.05$).

A non-compartment model was used to calculate the pharmacokinetic parameters of each group due to the extravascular absorption (Table 2). The effective half-lives of propofol in blood from mice pre-treated with (-)-borneol, (-)-carveol and *trans*-sobrerol groups were 21, 41 and 34 min, respectively whereas the half-life of propofol in the control group was 33 min. The terminal half-life of the (-)-borneol pre-treated group was about twice that of the control group. The change in total clearance (CL_T) of propofol from the blood in mice pre-treated with (-)-borneol, (-)-carveol, and *trans*-sobrerol was not significant. Mean residence time (MRT) was not significantly different between the control and the three groups pre-treated with MA. The volume of distribution at steady state (V_{ss}) was significantly lower only in the (-)-borneol pre-treated group ($p<0.05$). The maximum propofol concentration (C_{max}) in the (-)-borneol pre-treated group was also significantly higher than the propofol control group ($P<0.001$). The area under the curve (AUC) was significantly larger in the groups pre-treated with (-)-borneol and (-)-carveol as compared to the propofol control.

Inhibition of propofol metabolism by (-)-borneol, (-)-carveol, or trans-sobrerol

We also studied the inhibitory effects of (-)-borneol, (-)-carveol, and *trans*-sobrerol towards propofol glucuronide formation in isolated hepatocytes. Fig. 5 showed that (-)-carveol, *trans*-

sobrerol, and (-)-borneol inhibited hepatocyte propofol metabolism. (-)-Borneol was the most effective at decreasing propofol glucuronide formation ($P<0.05$). As shown in Table 3A, borneol at 200 μ M, significantly inhibited microsomal glucuronidation of propofol (200 μ M) ($P<0.05$). On the other hand, (-)-carveol and *trans*-sobrerol did not inhibit propofol glucuronidation.

Inhibition of aminopyrine-N-demethylation by propofol, (-)-borneol, (-)-carveol, menthol, and trans-sobrerol

As propofol (Guitton et al., 1998; Niwa et al., 1999) and aminopyrine (Niwa et al., 1999) have a similar P450 isozyme specificity (CYP2B and CYP2C) with little activity for CYP3A and CYP2E1, the effects of MAs on aminopyrine *N*-demethylation was tested. As shown in Table 3B, menthol, (-)-carveol, and *trans*-sobrerol significantly inhibited aminopyrine-*N*-demethylation activity preincubated with the microsomes ($P<0.001$). However, (-)-borneol at 0.5 mM did not affect CYP catalyzed metabolism.

Discussion

Humans are exposed to MAs as they are ingredients in food, cosmetics and herbal medicinal products. The interactions of anesthetics are not known. The mouse study described in this paper shows that ($\pm$)-borneol, (-)-carveol, *trans*-sobrerol, and (-)-menthol enhanced the anesthetic effect of propofol.

The blood propofol concentration-time profile corresponded to a three-compartment open model and showed an initial decline associated with redistribution and a second decline associated with rapid clearance by metabolism (Simons et al., 1991). A pharmacokinetic study showed that the blood concentration of propofol in CD-1 male mice reached 0.17 mM after a 100 mg/kg propofol administration. The propofol concentration in the mouse blood reached 0.18–0.33 mM when mice were pre-treated with the glucuronidation inhibitor (-)-borneol or the P450 inhibitors (-)-carveol, and *trans*-sobrerol. This indicates an increase of propofol concentration in the mouse blood. The (-)-borneol pre-treated group showed a significant decrease in V_{ss} , and an increase in C_{max} , and a possible decrease in CL_T of propofol. The pharmacokinetic data of *trans*-sobrerol and (-)-carveol pre-treated groups also showed similar trends as the (-)-borneol group. The increase in blood propofol levels could be explained if (-)-borneol, (-)-carveol, and *trans*-sobrerol primarily inhibited propofol metabolism and thereby increased propofol concentration in the blood. Previous evidence has suggested that propofol is not cleared without undergoing further metabolism either via CYP catalysed metabolism or phase II metabolizing enzymes (Le Guellec et al., 1995). Moreover, MAs alone did not induce any anesthetic effects in the mice, thereby excluding the possibility of the pharmacodynamic effect of the MAs alone. Therefore, it is likely that the anesthetic-prolonging effect of propofol is primarily due to the inhibition of its metabolism.

The protein binding of propofol is approximately 96–98% (Cockshott et al., 1992). Studies have shown that 50% of the

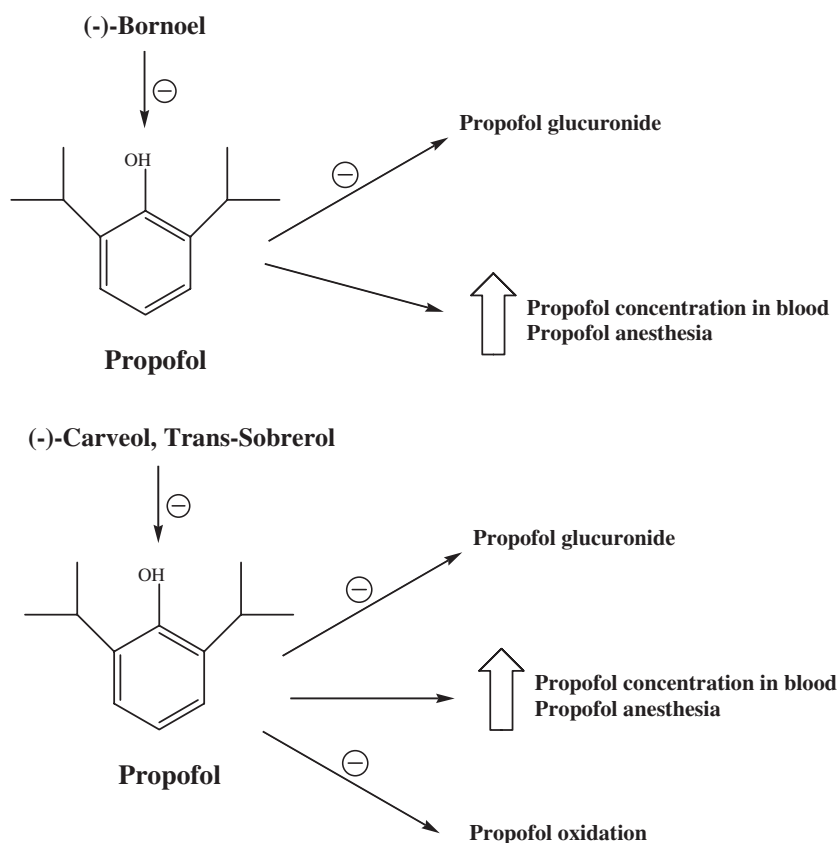
propofol is bound to erythrocytes, 48% to serum albumin leaving only 1.2% to 1.7% of unbound, free fraction remaining in the plasma (Mazoit and Samii, 1999). Therefore, the rate of propofol metabolism through glucuronic acid conjugation may be partially limited by its extravasation into the liver parenchyma (Mazoit and Samii, 1999). The interaction of MAs with propofol binding sites in blood components would likely increase the availability of propofol for glucuronidation. Although the effect of this possible interaction between MAs and propofol was not studied here, it could also play a role. Potentially, MAs could cause displacement of propofol from protein binding sites and increase the free fraction of propofol which would cause an increase in volume of distribution resulting in a decrease in blood propofol concentration and increase the hepatic metabolism of propofol. However, our pharmacokinetic data showed that borneol, (–)-carveol, and *trans*-sobrerol decreased the volume of distribution of propofol. The observed increase in propofol blood concentration together with the pharmacokinetic data strongly suggest that the MAs acted by inhibiting propofol metabolism.

(–)-Borneol, (–)-carveol, and *trans*-sobrerol decreased the propofol metabolic rate but only (–)-borneol decreased hepatocyte propofol glucuronide formation. On the other hand, (–)-carveol and *trans*-sobrerol inhibited propofol metabolism likely by inhibiting P450 catalysed hydroxylation without inhibiting propofol glucuronidation formation. Previous findings have shown that borneol is a potent inhibitor of liver microsomal glucuronidation (Gelal et al., 1999; Li et al., 2003). The lack of

inhibition of aminopyrine metabolism by (–)-borneol suggests that borneol is a poor P450 substrate or inhibitor and that borneol inhibits propofol metabolism by inhibiting glucuronidation. Therefore, it is clear that (–)-borneol increases the anesthetic effect of propofol solely by inhibiting propofol glucuronidation. Borneol acts by depleting hepatic UDPGA concentrations in vivo and can also act as a competitive inhibitor of UGT. Of 16 monoterpenoids and aliphatic alcohol substrates tested borneol was the best substrate for UTG2B12 (Green et al., 1995).

The major P450 isozymes that catalyze propofol oxidation or aminopyrine *N*-demethylation are CYP2B and CYP2C (Niwa et al., 1999). Microsomal aminopyrine *N*-demethylation was inhibited by (–)-carveol and *trans*-sobrerol but not (–)-borneol. These results suggest that (–)-carveol and *trans*-sobrerol increased the anesthetic effect of propofol as a result of their inhibition of P450. (–)-Carveol and *trans*-sobrerol have been shown to undergo P450 catalyzed oxidation to rat and human liver microsomes (Ventura et al., 1983, 1985) and would be expected to inhibit propofol metabolism by competitive metabolism.

In conclusion, using in vivo and in vitro studies we have demonstrated that exposure to MAs can increase propofol concentration by inhibiting propofol metabolism. (–)-Borneol increased propofol concentration primarily by inhibiting propofol glucuronidation whereas (–)-carveol and *trans*-sobrerol decreased propofol metabolism by inhibiting oxidation and glucuronidation (Scheme 1). It is likely that dietary monoterpene would interact with other drugs. Better understanding of



Scheme 1. The proposed mechanism of the inhibition of propofol metabolism by (–)-borneol, (–)-carveol, and *trans*-sobrerol in different pathways.

these interactions, particularly at doses more relative to those consumed by humans, will require additional research.

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